Advanced FEM Techniques (AFEMT)

Lecture 2

Andrea Cangiani

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(Conforming) Galerkin method

Consider the slightly more general problem:

$$-d\Delta u = f \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega.$$

with given diffusion coefficient $d = d(x)$.

Let V_h a continuous piecewise polynomial space with respect to a partition (mesh) of the domain.

Conforming (continuous) FEM:Find $u_h \in V_h$:

$$\int_\Omega d(x)\,\nabla u_h(x) \cdot \nabla v(x)\,dx = \int_\Omega f(x)\,v(x)\,dx \quad \forall v \in V_h.$$

FEM Triplet

A finite element is described by the triplet [Ciarlet, SIAM, 2002]

$$(K, P, \Sigma),$$

where

- K is the cell,
- P is the local space of local shape functions (polynomials)

$$\phi : K \to \mathbb{R}^d,$$

  of dimension n_{sh}.
- $\Sigma = \{\sigma_i\}_{i=1}^{n_{sh}}$ set of unisolvent[1] linear forms (degrees of freedom) acting on P

[1] The linear map $\Phi(p) = (\sigma_1(p), \ldots, \sigma_{n_{sh}}(p))$ is bijective.

Lagrange elements in deal.ii

Obviously, there exists a basis $\{\phi_j\}_{j=1}^{n_{sh}}$ determined by the property that

$$\sigma_i(\phi_j) = \delta_{ij}.$$

When all σ_i are point evaluations we call this a Lagrange basis.

Main FE in deal.II

- $\hat{K} = [0,1]^d$
- $\hat{P} = \mathbb{Q}_k^{(d)} := \mathrm{span}\{x_1^{\beta_1} \cdots x_d^{\beta_d}, \quad 0 \leq \beta_1, \ldots, \beta_d \leq k\}$ with basis
  - $d = 1$: Gauss-Lobatto functions
  - $d > 1$: tensor product of one-dimensional functions
- $\hat{\Sigma}$ the corresponding tensor product nodal (Lagrange) basis

deal.II corrispondence of concepts

- mesh cells $\longleftrightarrow$ `Triangulation`
- shape functions $\longleftrightarrow$ `FEValues`, ...
- degrees of freedom $\longleftrightarrow$ `DoFHandler`

Mesh in deal.II

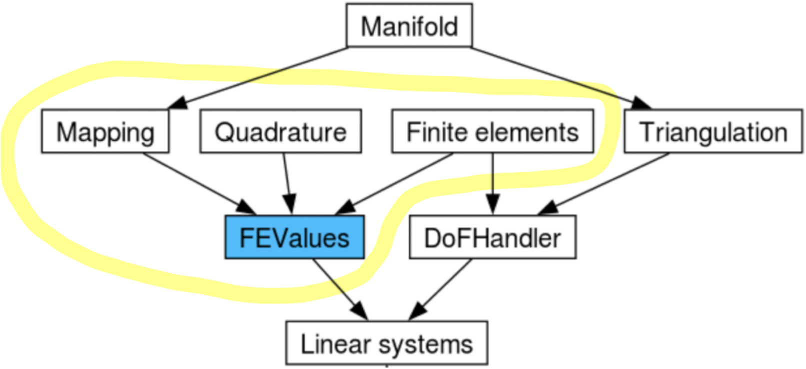
- `Triangulation` stores cells, vertices, faces and connectivity.
- `GridGenerator` provides standard meshes.
- Refinement acts directly on the triangulation.

Example

```
Triangulation<2> triangulation;
GridGenerator::hyper_cube(triangulation, 0.0, 1.0);
triangulation.refine_global(2);
```

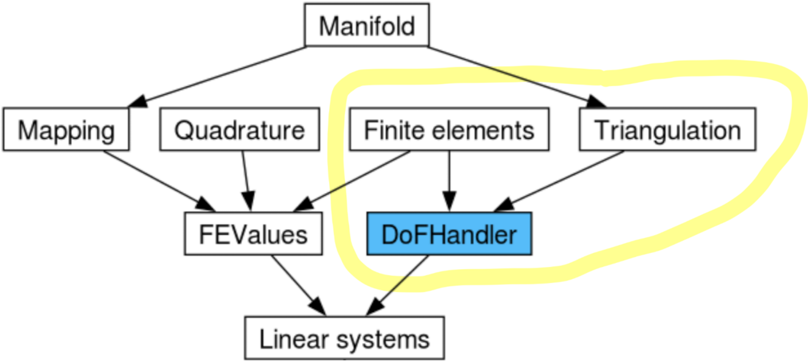
deal.ii architecture: `FiniteElement`, `FEValues`

`FiniteElement`: abstract base class for FE in any dimension.



`FEValues` links the `FiniteElement`, `quadrature`, and `mapping` classes to produce the required values of the physical element's shape functions

deal.ii architecture: DoFHandler



DoFHandler given a FiniteElement and a Triangulation enumerates the degrees of freedom.[2]

Note! the nodes (vertices) of the triangulation ar in general different from the degrees of freedom.

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[2] It allows queries of DoF as well as the triangulation data.

Classical Continuous Elements

For conforming methods,

$$V_h \subset H^1(\Omega).$$

A prototype continuous space on a quadrilateral mesh is

$$V_h = \{v_h \in C^0(\Omega) : v_h|_K \circ T_K \in \mathbb{Q}_k(\hat{K}),\ \forall K \in \mathcal{T}_h\}.$$

Theorem

For $v \in H^1(\mathcal{T}_h)$ we have

$$v \in H^1(\Omega) \quad \Leftrightarrow \quad jump(v)|_F = 0,\ \forall F \text{ internal face of } \mathcal{T}_h.$$

Continuity is enforced by identifying interface degrees of freedom.
In `deal.ii` for, say `FE_Q`, this is done automaticlly exploiting the fact that elemental interface values are determined by the iterfaces DoF and applying interface contraints. Same for boundary values.

Linear System

Write $u_h = \sum_{j=1}^{N} U_j \phi_j$. Testing with basis functions gives

$$\sum_{j=1}^{N} U_j a(\phi_j, \phi_i) = F(\phi_i), \qquad i = 1, \ldots, N.$$

Hence

$$AU = F, \qquad A_{ij} = a(\phi_j, \phi_i), \qquad F_i = F(\phi_i).$$

Now, note that

$$A_{ij} = \sum_{K \subset \mathrm{supp}(\phi_i) \cap \mathrm{supp}(\phi_j)} \underbrace{\int_K d(x) \nabla \phi_j(x) \cdot \nabla \phi_i(x)\, dx}_{A_{ij}^K}$$

DoFHandler

$$A_{ij} = \sum_{K \subset \mathrm{supp}(\phi_i) \cap \mathrm{supp}(\phi_j)} \underbrace{\int_K d(x) \nabla \phi_j(x) \cdot \nabla \phi_i(x)\, dx}_{A_{ij}^K}$$

Crucial idea: FEM built from local objects + global connectivity.

$$\boxed{\text{cell } K} \xrightarrow{\text{quadrature}} \boxed{\begin{array}{c}\text{local matrix } A^K\\ \text{local rhs } F^K\end{array}} \xrightarrow{\text{DoF indices}} \boxed{\begin{array}{c}\text{global system}\\ AU = F\end{array}}$$

`DoFHandler` attaches global indices to local degrees of freedom.

$$\text{local basis on a cell} \quad \longrightarrow \quad \text{global numbering}$$

Bridge between cell-wise calculations and the global linear system.

Assembly

To assemble the stiffness matrix, we need to compute on each element, for $i, j \in \{1, \ldots, n_{sh}\}$ terms like

$$\int_K d(x)\, \nabla\phi_j(x) \cdot \nabla\phi_i(s)\, dx$$

$$= \int_{\hat{K}} d(T_K(\hat{x}))\, J_K^{-T} \nabla_{\hat{x}}\hat{\phi}_j(\hat{x}) \cdot J_K^{-T} \nabla_{\hat{x}}\hat{\phi}_i(\hat{x})\, |\det J_K(\hat{x})|\, d\hat{x}$$

$$\approx \sum_{l=1}^{n_{qp}} d(T_K(\hat{\xi}_l))\, J_K^{-T} \nabla_{\hat{x}}\hat{\phi}_j(\hat{\xi}_l) \cdot J_K^{-T} \nabla_{\hat{x}}\hat{\phi}_i(\hat{\xi}_l)\, |\det J_K(\hat{\xi}_l)|\, \hat{w}_l$$

where $\{\hat{\xi}_l, , \hat{w}_l\}_{l=1}^{n_{qp}}$ is the set of quadrature points and weights and J_K is the Jacobian of T_K.

- geometry and ref. shape functions coupled through mapping.
- choose the quadrature order according to the degree of the integrand (under-integration can destroy accuracy).

FEValues (Core Object)

At quadrature points, `FEValues` provides

- values $\phi_i(x_q)$,
- gradients $\nabla\phi_i(x_q)$,
- weighted Jacobians `JxW(q)`.

In other words, it turns the abstract bilinear form into something computable.

A priori error analysis

Theorem (Lemma of Cèa)

$$|u - u_h|_{1,\Omega} \lesssim \inf_{v \in V_h} |u - v|_{1,\Omega}$$

Theorem (Interpolation error)

Let I_h be the local nodal interpolation operator.
On shape-regular meshes, for sufficiently smooth u,

$$\|u - I_h u\|_{H^m(K)} \leq C h_K^{k+1-m} |u|_{H^{k+1}(K)}, \qquad 0 \leq m \leq k+1.$$

and, globally,

$$\|u - I_h u\|_{H^1(\Omega)} \leq C h^k |u|_{H^{k+1}(\Omega)}.$$

$\Rightarrow$ resulting FEM expected to be of order k in energy norm.

Quadrature error

Theorem (Ciarlet, Chapter 4 [Ciarlet, SIAM, 2002])

The order of the FEM is preserved if quadrature is exact on all polynomials of degree $2k - 2$.

In `deal.ii`, the Gauss–Legendre rules `QGauss` are used. With n quadrature point, this is exact on $2n - 1$. Hence, in this case, $n = k$ is enough.

We shall use $n = k + 1$. For instance, this guarantees optimality in L^2-norm also.

deal.II Assembly Code: local contributions

```
for (const auto &cell : dof_handler.active_cell_iterators())
{
  fe_values.reinit(cell);
  cell_matrix = 0;
  cell_rhs = 0;

  for (unsigned int q=0; q<n_q_points; ++q)
    for (unsigned int i=0; i<dofs_per_cell; ++i)
    {
      for (unsigned int j=0; j<dofs_per_cell; ++j)
        cell_matrix(i,j) += fe_values.shape_grad(i,q) *
                            fe_values.shape_grad(j,q) *
                            fe_values.JxW(q);

      cell_rhs(i) += rhs_value(q) *
                     fe_values.shape_value(i,q) *
                     fe_values.JxW(q);
    }
```

deal.II Assembly Code: local-to-global

```
cell->get_dof_indices(local_dof_indices);

for (const unsigned int i : fe_values.dof_indices())
   for (const unsigned int j : fe_values.dof_indices())
      system_matrix.add(local_dof_indices[i],
                        local_dof_indices[j],
                        cell_matrix(i, j));

 for (const unsigned int i : fe_values.dof_indices())
   system_rhs(local_dof_indices[i]) += cell_rhs(i);
```

Common Pitfalls

- confusing nodes with quadrature points,
- using too low a quadrature order,
- forgetting that `FEValues` is local while `DoFHandler` is global,
- ignoring constraints when assembling the global system.